

Contoso

MODULE 4

SUBTITLE COMES HERE

MODULE 4

- Introduction, Meaning Representation
- Lexical Semantics
- Corpus Study, Study of Various language dictionaries like wordnet, Babelnet
- Relations among lexemes & their senses –Homonymy, Polysemy, Synonymy, Hyponymy
- Semantic Ambiguity
- Word Sense Disambiguation (WSD)
- Knowledge based approach (Lesk's Algorithm)
- Supervised (Naïve Bayes, Decision List)
- Introduction to Semi-supervised method (Yarowsky) Unsupervised (Hyperlex)

SEMANTIC ANALYSIS

- Semantic analysis is the process of finding the meaning from text.

- This analysis gives the power to computers
  - to understand and interpret sentences, paragraphs, or whole documents,
  - by analyzing their grammatical structure
  - identifying the relationships between individual words of the sentence in a particular context.

LEXICAL SEMANTICS

- It is the study of the meaning of individual words.
- It involves words, sub-words, affixes (sub-units), compound words, and phrases also.
- All the words, sub-words, etc. are collectively known as lexical items.

Three levels of meaning

1. Lexical Semantics (words) 
2. Sentential / Compositional / Formal Semantics
3. Discourse or Pragmatics
   - meaning + context + world knowledge

Terminology

- **Lexeme:** a pairing of meaning and form
- **Lemma:** the word form that represents a **lexeme**
  - ***Carpet*** is the lemma for ***carpets***
  - ***Dormir*** is the lemma for ***duermes***

- The lemma *bank* has two **senses:**
  - *Financial insitution*
  - *Soil wall next to water*

- A **sense** is a discrete representation of one aspect of the meaning of a word

Lemmas have senses

- One lemma "bank" can have many meanings:

Sense 1: • …a **bank**$_1$ can hold the investments in a custodial account…

Sense 2: • "…as agriculture burgeons on the east **bank**$_2$ the river will shrink even more"

- **Sense** (or **word sense**)
  - A discrete representation of an aspect of a word's meaning.
- The lemma **bank** here has two senses

Relations between words/senses

- Homonymy
- Polysemy
- Synonymy
- Antonymy
- Hypernymy
- Hyponymy
- Meronymy

Polysemy

- 1. The **bank** was constructed in 1875 out of local red brick.
- 2. I withdrew the money from the **bank**
- Are those the same sense?
  - Sense 2: “A financial institution”
  - Sense 1: “The building belonging to a financial institution”
- A **polysemous** word has **related** meanings
  - Most non-rare words have multiple meanings

Trying to make sense of senses

Polysemy:

A lexeme is polysemous if it has different *related senses*





bank = financial institution or building

Homonyms:

Two lexemes are homonyms if their *senses are unrelated*, but they happen to have the **same spelling and pronunciation**





bank = (financial) bank or (river) bank

Metonymy or Systematic Polysemy: A systematic relationship between senses

- Lots of types of polysemy are systematic
  - `School, university, hospital`
  - All can mean the institution or the building.
- A systematic relationship:
  - Building ⟷ Organization
- Other such kinds of systematic polysemy:

Author (`Jane Austen wrote Emma`)

⟷ Works of Author (`I love Jane Austen`)

Tree (`Plums have beautiful blossoms`)

⟷ Fruit (`I ate a preserved plum`)

POLYSEMY

Multiple related meanings

S: (n) **newspaper**, paper (a daily or weekly publication on folded sheets; contains news and articles and advertisements) *"he read his newspaper at breakfast"*

S: (n) **newspaper**, paper, newspaper publisher (a business firm that publishes newspapers) *"Murdoch owns many newspapers"*

S: (n) **newspaper**, paper (the physical object that is the product of a newspaper publisher) *"when it began to rain he covered his head with a newspaper"*

S: (n) **newspaper**, newsprint (cheap paper made from wood pulp and used for printing newspapers) *"they used bales of newspaper every day"*

How do we know when a word has more than one sense?

- The "zeugma" test: Two senses of `serve`?
  - `Which flights **serve** breakfast?`
  - `Does Lufthansa **serve** Philadelphia?`
  - Does Lufthansa serve breakfast and San Jose?
- Since this conjunction sounds weird,
  - we say that these are **two different senses of "serve"**

Homonymy

- Homonyms: lexemes that share a form, but unrelated meanings
- Examples:
  - *bat* (wooden stick thing) vs *bat* (flying scary mammal)
  - *bank* (financial institution) vs *bank* (riverside)
- Can be homophones, homographs, or both:
  - Homophones: *write* and *right*, *piece* and *peace*
  - Homographs: *bass* and *bass*

Homonymy: unrelated Polysemy: related meaning

S: (n) **position**, place (the particular portion of space occupied by something) *"he put the lamp back in its place"*

S: (n) military position, **position** (a point occupied by troops for tactical reasons)

S: (n) **position**, view, perspective (a way of regarding situations or topics etc.)*"consider what follows from the positivist view"*

S: (n) **position**, posture, attitude (the arrangement of the body and its limbs) *"he assumed an attitude of surrender"*

S: (n) status, **position** (the relative position or standing of things or especially persons in a society) *"he had the status of a minor"; "the novel attained the status of a classic"; "atheists do not enjoy a favorable position in American life"*

S: (n) **position**, post, berth, office, spot, billet, place, situation (a job in an organization) *"he occupied a post in the treasury"*

Homonymy causes problems for NLP applications

- Information retrieval
  - "`bat care`"
- Machine Translation
  - `bat:` murciélago (animal) or bate (for baseball)
- Text-to-Speech
  - `bass` (stringed instrument) vs. `bass` (fish)

Relations between senses

Symmetric relations:

Synonyms: *couch/sofa*

Two lemmas with the **same** sense

Antonyms: *cold/hot, rise/fall, in/out*

Two lemmas with the **opposite** sense

Hierarchical relations:

Hypernyms and **hyponyms**: *pet/dog*

The hyponym *(dog)* is **more specific** than the hypernym *(pet)*

Holonyms and **meronyms**: *car/wheel*

The meronym *(wheel)* is a **part of** the holonym *(car)*

Synonyms

- Words that have the same meaning in some or all contexts.
  - filbert / hazelnut
  - couch / sofa
  - big / large
  - automobile / car
  - vomit / throw up
  - Water / H_20
- Two words are synonyms if they can be substituted for each other in all situations (strict/perfect definition).

Synonyms

- But there are few (or no) examples of perfect synonymy.
  - Even if many aspects of meaning are identical
  - Still may not preserve the acceptability based on notions of politeness, slang, register, genre, etc.
- Example:
  - Water/H_20
  - Big/large
  - Brave/courageous

Synonymy is a relation between senses rather than words

- Consider the words *big* and *large*
- Are they synonyms?
  - How **big** is that plane?
  - Would I be flying on a **large** or small plane?
- How about here:
  - Miss Nelson became a kind of **big** sister to Benjamin.
  - Miss Nelson became a kind of **large** sister to Benjamin.
- Why?
  - *big* has a sense that means being older, or grown up
  - *large* lacks this sense

Antonyms

- Senses that are opposites with respect to one feature of meaning
- Otherwise, they are very similar!

  `dark/light` `short/long` `fast/slow` `rise/fall`

  `hot/cold` `up/down` `in/out`

- More formally: antonyms can
  - define a binary opposition or be at opposite ends of a scale
    - `long/short, fast/slow`
  - Be **reversives**:
    - `rise/fall, up/down`

Hyponyms and Hypernyms

- **Hyponym**: the sense is a subclass of another sense
  - *car* is a hyponym of *vehicle*
  - *dog* is a hyponym of *animal*
  - *mango* is a hyponym of *fruit*
- **Hypernym**: the sense is a superclass
  - *vehicle* is a hypernym of *car*
  - *animal* is a hypernym of *dog*
  - *fruit* is a hypernym of *mango*

hypernym	vehicle	fruit	furniture	mammal
hyponym	car	mango	chair	dog

QUIZ

Classify the following examples in terms of what lexical semantic relation they exhibit

cold	*freezing*
they're	*their*
hair	*head*
enemy	*friend*
cut (hair)	*cut (bread)*
George Clooney	*actor*

WORDNET

Very large lexical database of English:

110K nouns, 11K verbs, 22K adjectives, 4.5K adverbs

(WordNets for many other languages exist or are under construction)

Word senses grouped into synonym sets (“synsets”) linked into a conceptual-semantic hierarchy

81K noun synsets, 13K verb synsets, 19K adj. synsets, 3.5K adv synsets Avg. # of senses: 1.23 nouns, 2.16 verbs, 1.41 adj, 1.24 adverbs

Conceptual-semantic relations: hypernym/hyponym

also holonym/meronym

Also lexical relations, in particular lemmatization

Available at http://wordnet.princeton.edu

A WORDNET EXAMPLE

WordNet Search - 3.0 - WordNet home page - Glossary - Help

Word to search for: bass Search WordNet

Display Options: (Select option to change) Change

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

Noun

- S: (n) **bass** (the lowest part of the musical range)
- S: (n) **bass**, bass part (the lowest part in polyphonic music)
- S: (n) **bass**, basso (an adult male singer with the lowest voice)
- S: (n) sea bass, **bass** (the lean flesh of a saltwater fish of the family Serranidae)
- S: (n) freshwater bass, **bass** (any of various North American freshwater fish with lean flesh (especially of the genus Micropterus))
- S: (n) **bass**, bass voice, basso (the lowest adult male singing voice)
- S: (n) **bass** (the member with the lowest range of a family of musical instruments)
- S: (n) **bass** (nontechnical name for any of numerous edible marine and freshwater spiny-finned fishes)

Adjective

- S: (adj) **bass**, deep (having or denoting a low vocal or instrumental range) *"a deep voice"; "a bass voice is lower than a baritone voice"; "a bass clarinet"*

WordNet home page

HIERARCHICAL SYNSET RELATIONS: NOUNS

Hypernym/hyponym (between concepts)
The more general '***meal'*** is a hypernym of the more specific '***breakfast'***

Instance hypernym/hyponym (between concepts and instances)
Austen is an instance hyponym of ***author***

Member holonym/meronym (groups and members)
professor is a member meronym of (a university's) ***faculty***

Part holonym/meronym (wholes and parts)
wheel is a part meronym of (is a part of) ***car.***

Substance meronym/holonym (substances and components)
flour is a substance meronym of (is made of) ***bread***

HIERARCHICAL SYNSET RELATIONS: VERBS

Hypernym/troponym (between events):
travel/fly, walk/stroll
Flying is a troponym of *traveling:*
it denotes **a specific manner** of *traveling*

Entailment (between events):
snore/sleep
Snoring **entails (presupposes)** *sleeping*

WORDNET HYPERNYMS AND HYPONYMS

- S: (n) **bass** (the lowest part of the musical range)
  - *direct hypernym* / *inherited hypernym* / *sister term*
    - S: (n) pitch (the property of sound that varies with variation in the frequency of vibration)
      - S: (n) sound property (an attribute of sound)
        - S: (n) property (a basic or essential attribute shared by all members of a class) "*a stuc*
          - S: (n) attribute (an abstraction belonging to or characteristic of an entity)
            - S: (n) abstraction, abstract entity (a general concept formed by extracting
              - S: (n) entity (that which is perceived or known or inferred to have
- S: (n) **bass**, bass part (the lowest part in polyphonic music)
  - *direct hyponym* / ***full hyponym***
    - S: (n) ground bass (a short melody in the bass that is constantly repeated)
    - S: (n) figured bass, basso continuo, continuo, thorough bass (a bass part written out in full and acc
  - *direct hypernym* / *inherited hypernym* / *sister term*
    - S: (n) part, voice (the melody carried by a particular voice or instrument in polyphonic music) "*he*
      - S: (n) tune, melody, air, strain, melodic line, line, melodic phrase (a succession of notes forı
        - S: (n) music (an artistic form of auditory communication incorporating instrumental
          - S: (n) auditory communication (communication that relies on hearing)
            - S: (n) communication (something that is communicated by or to or betw
              - S: (n) abstraction, abstract entity (a general concept formed by ext
                - S: (n) entity (that which is perceived or known or inferred to

WORD SENSE DISAMBIGUATION

WORD SENSE DISAMBIGUATION

What does this word mean?

This **plant** needs to be watered each day.

⇒ living plant

This **plant** manufactures 1000 widgets each day.

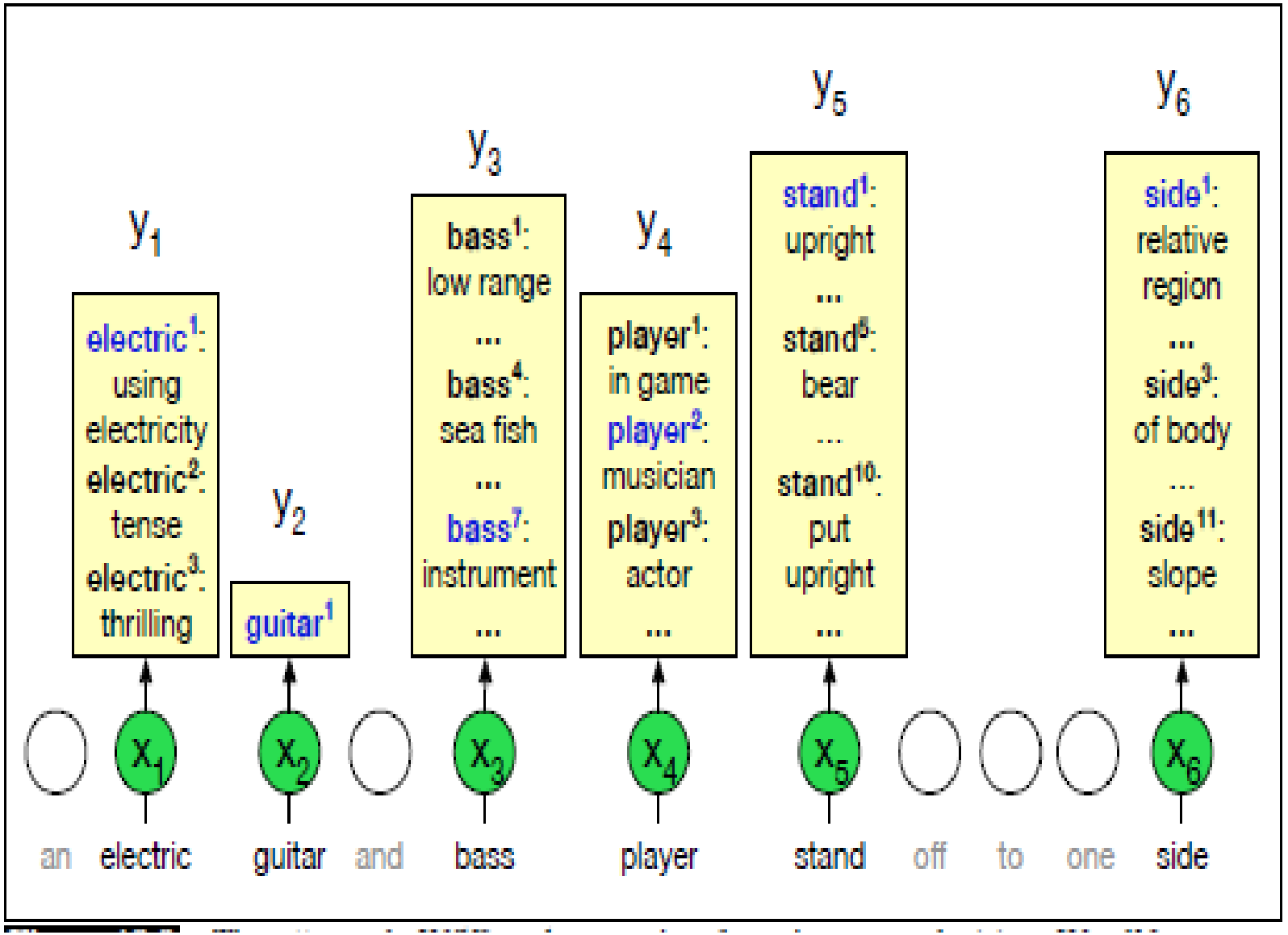
⇒ factory

WORD SENSE DISAMBIGUATION

- The task of selecting the correct sense for a word is called word sense disambiguation, or WSD. word sense.
- Word Sense Disambiguation is an important method of NLP by which the meaning of a word is determined, which is used in a particular context.
- NLP systems often face the challenge of properly identifying words, and determining the specific usage of a word in a particular sentence.
- Word Sense Disambiguation basically solves the ambiguity that arises in determining the meaning of the same word used in different situations.

THE DATA

Sense	Training Examples (Keyword in Context)
?	... company said the *plant* is still operating
?	Although thousands of *plant* and animal species
?	... zonal distribution of *plant* life . ...
?	... to strain microscopic *plant* life from the ...
?	vinyl chloride monomer *plant* , which is ...
?	and Golgi apparatus of *plant* and animal cells
?	... computer disk drive *plant* located in ...
?	... divide life into *plant* and animal kingdom
?	... close-up studies of *plant* life and natural
?	... Nissan car and truck *plant* in Japan is ...
?	... keep a manufacturing *plant* profitable without
?	... molecules found in *plant* and animal tissue
?	... union responses to *plant* closures . ...
?	... animal rather than *plant* tissues can be
?	... many dangers to *plant* and animal life
?	company manufacturing *plant* is in Orlando ...
?	... growth of aquatic *plant* life in water ...
?	automated manufacturing *plant* in Fremont ,
?	... Animal and *plant* life are delicately
?	discovered at a St. Louis *plant* manufacturing
?	computer manufacturing *plant* and adjacent ...
?	... the proliferation of *plant* and animal life
?	... ...

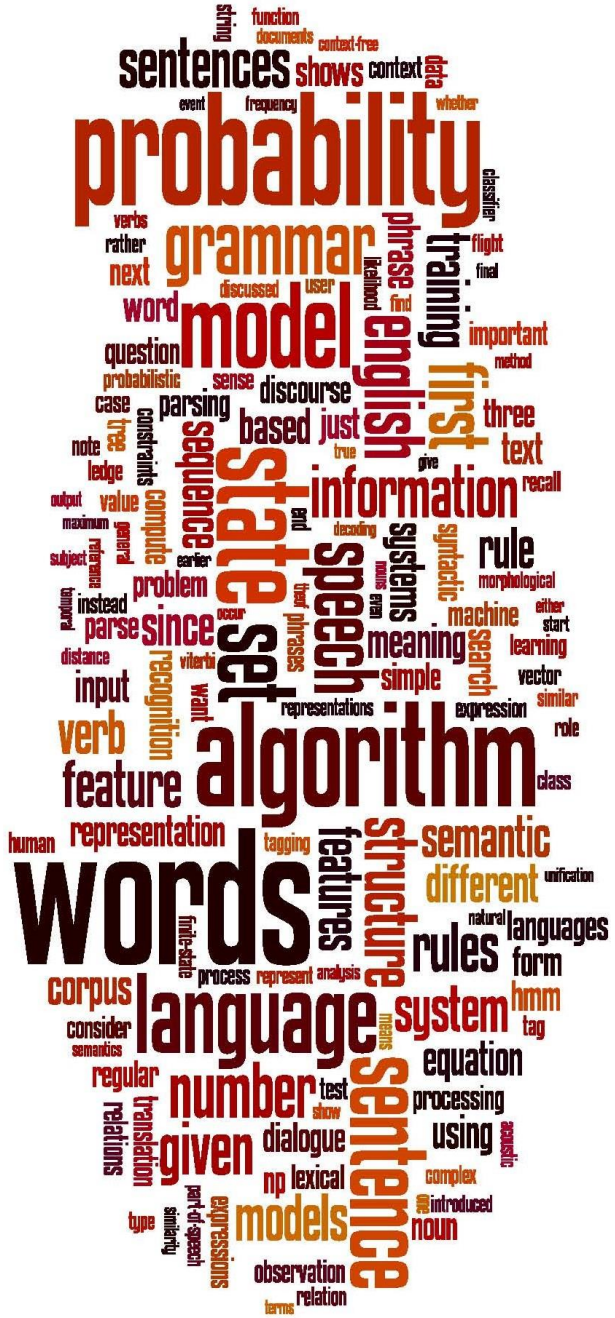

y1
electric1:
using
electricity
electric2:
tense
electric3:
thrilling
y2
guitar1
y3
bass1:
low range
...
bass4:
sea fish
...
bass7:
instrument
...
y4
player1:
in game
player2:
musician
player3:
actor
...
y5
stand1:
upright
...
stand8:
bear
...
stand10:
put
upright
...
y6
side1:
relative
region
...
side3:
of body
...
side11:
slope
...
x1
x2
x3
x4
x5
x6
an
electric
guitar
and
bass
player
stand
off
to
one
side

LET US TAKE THE EXAMPLE OF THE WORD "BARK":

- One meaning of the word refers to the outer covering of the tree.
- The other meaning refers to the sound made by a dog. here the same word has different meanings.
- **Let us take a piece of text:**
- *"Cinnamon comes from the bark of the Cinnamon tree."*
- *"The dog barked at the stranger."*
- **Let us now try a sentence with both words:**
- *"The dog was scratching the bark of the tree, when the man approached the dog to make it stop, the dog barked."*
- if this sentence is passed to an algorithm for sentiment analysis, "*bark*" and "*barked*" might mean the same meaning.
- Hence, same words can mean differently based on the usage of the word in a particular sentence.

WSD METHODS

- Supervised Machine Learning
- Thesaurus/Dictionary Methods
- Semi--Supervised Learning



Word Sense Disambiguation

- Supervised Machine Learning

SUPERVISED MACHINE LEARNING APPROACHES

- Supervised machine learning approach:
  - a training corpus of words tagged in context with their sense
  - used to train a classifier that can tag words in new text
- Summary of what we need:
  - the **tag set** (“sense inventory”)
  - the **training corpus**
  - A set of **features** extracted from the training corpus
  - A **classifier**

SUPERVISED WSD 1: WSD TAGS

- What's a tag?

  A dictionary sense?

- For example, for WordNet an instance of "bass" in a text has 8 possible tags or labels (bass1 through bass8).

8 SENSES OF "BASS" IN WORDNET

1. bass –(the lowest part of the musical range)
2. bass, bass part –(the lowest part in polyphonic music)
3. bass, basso –(an adult male singer with the lowest voice)
4. sea bass, bass –(flesh of lean--fleshed saltwater fish of the family Serranidae)
5. freshwater bass, bass –(any of various North American lean--fleshed freshwater fishes especially of the genus Micropterus)
6. bass, bass voice, basso –(the lowest adult male singing voice)
7. bass –(the member with the lowest range of a family of musical instruments)
8. bass –(nontechnical name for any of numerous edible marine and freshwater spiny--finned fishes)

INVENTORY OF SENSE TAGS FOR *BASS*

WordNet Sense	Spanish Translation	Roget Category	Target Word in Context
bass^4	lubina	FISH/INSECT	. . . fish as Pacific salmon and striped **bass** and. . .
bass^4	lubina	FISH/INSECT	. . . produce filets of smoked **bass** or sturgeon. . .
bass^7	bajo	MUSIC	. . . exciting jazz **bass** player since Ray Brown. . .
bass^7	bajo	MUSIC	. . . play **bass** because he doesn't have to solo. . .

SUPERVISED WSD 2: GET A CORPUS

- Lexical sample task:
  - *Line–hard–serve* corpus -4000 examples of each
  - *Interest* corpus –2369 sense-tagged examples
- All words:
  - **Semantic concordance**: a corpus in which each open-class word is labeled with a sense from a specific dictionary/thesaurus.
    - SemCor: 234,000 words from Brown Corpus, manually tagged with WordNet senses
    - SENSEVAL--3 competition corpora -2081 tagged word tokens

SEMCOR

<wf pos=PRP>**He**</wf>

<wf pos=VB lemma=recognize wnsn=4 lexsn=2:31:00::>**recognized**</wf>

<wf pos=DT>**the**</wf>

<wf pos=NN lemma=gesture wnsn=1 lexsn=1:04:00::>**gesture**</wf>

<punc>.</punc>

SUPERVISED WSD 3: EXTRACT FEATURE VECTORS

INTUITION FROM WARREN WEAVER (1955):

"If one examines the words in a book, one at a time as through an opaque mask with a hole in it one word wide, then it is obviously impossible to determine, one at a time, the meaning of the words…

But if one lengthens the slit in the opaque mask, until one can see not only the central word in question but also say N words on either side, then if N is large enough one can unambiguously decide the meaning of the central word…

The practical question is : ``What minimum value of N will, at least in a tolerable fraction of cases, lead to the correct choice of meaning for the central word?"

FEATURE VECTORS

- A simple representation for each observation

  (each instance of a target word)

  - **Vectors** of sets of feature/value pairs
  - Represented as a ordered list of values
  - These vectors represent, e.g., the window of words around the target

TWO KINDS OF FEATURES IN THE VECTORS

- **Collocational** features and **bag–of–words** features
  - **Collocational**
    - Features about words at **specific** positions near target word
      - Often limited to just word identity and POS
  - **Bag–of–words**
    - Features about words that occur anywhere in the window (regardless of position)
      - Typically limited to frequency counts

EXAMPLES

- Example text (WSJ):

  An electric guitar and **bass** player stand off to one side not really part of the scene

- Assume a window of +/–2 from the target

EXAMPLES

- Example text (WSJ)

  An electric guitar and **bass** player stand off to one side not really part of the scene,

- Assume a window of +/–2 from the target

COLLOCATIONAL FEATURES

- Position-specific information about the words and collocations in window
- guitar and bass player stand

$$[w_{i-2}, \mathrm{POS}_{i-2}, w_{i-1}, \mathrm{POS}_{i-1}, w_{i+1}, \mathrm{POS}_{i+1}, w_{i+2}, \mathrm{POS}_{i+2}, w_{i-2}^{i-1}, w_{i}^{i+1}]$$

[guitar, NN, and, CC, player, NN, stand, VB, and guitar, player stand]

- word 1,2,3 grams in window of ±3 is common

BAG-OF-WORDS FEATURES

- “an unordered set of words” – position ignored
- Counts of words occur within the window.
- First choose a vocabulary
- Then count how often each of those terms occurs in a given window
  - sometimes just a binary “indicator” 1 or 0

CO--OCCURRENCE EXAMPLE

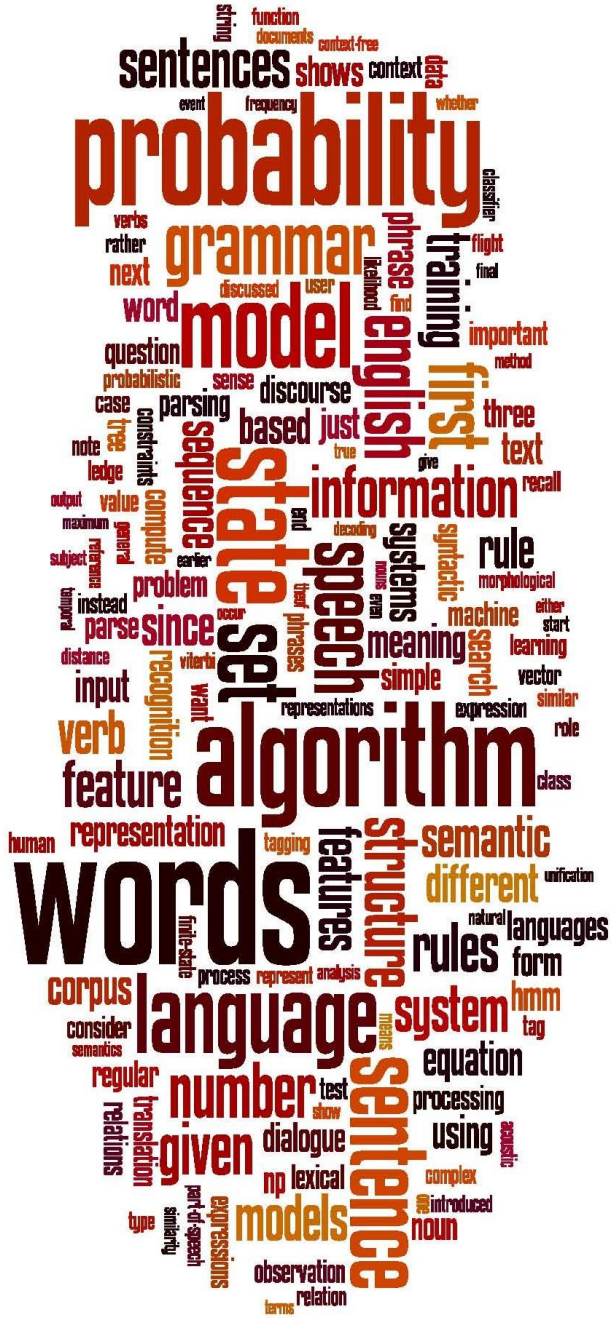
- Assume we've settled on a possible vocabulary of 12 words in "bass" sentences:

[*fishing, big, sound, player, fly, rod, pound, double, runs, playing, guitar, band*]

- The vector for:

  guitar and bass player stand

  [0,0,0,1,0,0,0,0,0,0,1,0]



Word Sense Disambiguation

Classification



CLASSIFICATION: DEFINITION

- *Input*:
  - a word w and some features *f*
  - a fixed set of classes $C = \{c_1, c_2, \ldots, c_J\}$

- *Output*: a predicted class $c \in C$



METHODS: SUPERVISED MACHINE LEARNING

- *Input:*
  - a word *w in a text window d (which we'll call a "document")*
  - a fixed set of classes $C = \{c_1, c_2, \ldots, c_J\}$
  - A training set of *m* hand--labeled text windows again called "documents" $(d_1,c_1),\ldots,(d_m,c_m)$
- *Output:*
  - a learned classifier $\gamma:d \rightarrow c$



METHODS: SUPERVISED MACHINE LEARNING

- Any kind of classifier
  - Naive Bayes
  - Logistic regression
  - Neural Networks
  - Support--vector machines
  - k--Nearest Neighbors
  - …

APPLYING NAIVE BAYES TO WSD

- P(c) is the prior probability of that sense
  - Counting in a labeled training set.
- P(w|c) conditional probability of a word given a particular sense
  - P(w|c) = count(w,c)/count(c)
- We get both of these from a tagged corpus like SemCor

- Can also generalize to look at other features besides words.
  - Then it would be P(f|c)
    - Conditional probability of a feature given a sense



$$\hat{P}(C) = \frac{N_C}{N}$$

$$\hat{P}(w \mid c) = \frac{count(w,c)+1}{count(c)+|V|}$$

	Doc	Words	Class
Training	1	fish smoked fish	f
	2	fish line	f
	3	fish haul smoked	f
	4	guitar jazz line	g
Test	5	line guitar jazz jazz	?

Priors:

$P(f) = \frac{3}{4}$

$P(g) = \frac{1}{4}$

V = {fish, smoked, line, haul, guitar, jazz}

Conditional Probabilities:

P(line|*f*) = (1+1) / (8+6) = 2/14

P(guitar|*f*) = (0+1) / (8+6) = 1/14

P(jazz|*f*) = (0+1) / (8+6) = 1/14

P(line|*g*) = (1+1) / (3+6) = 2/9

P(guitar|*g*) = (1+1) / (3+6) = 2/9

P(jazz|*g*) = (1+1) / (3+6) = 2/9

Choosing a class:

P(f|d5) $\propto$ 3/4 * 2/14 * $(1/14)^2$ * 1/14
≈ 0.00003

P(g|d5) $\propto$ 1/4 * 2/9 * $(2/9)^2$ * 2/9
≈ 0.0006

DECISION LISTS

A decision list is an ordered list of yes-no questions

***bass1* = fish vs. *bass2* = music:**

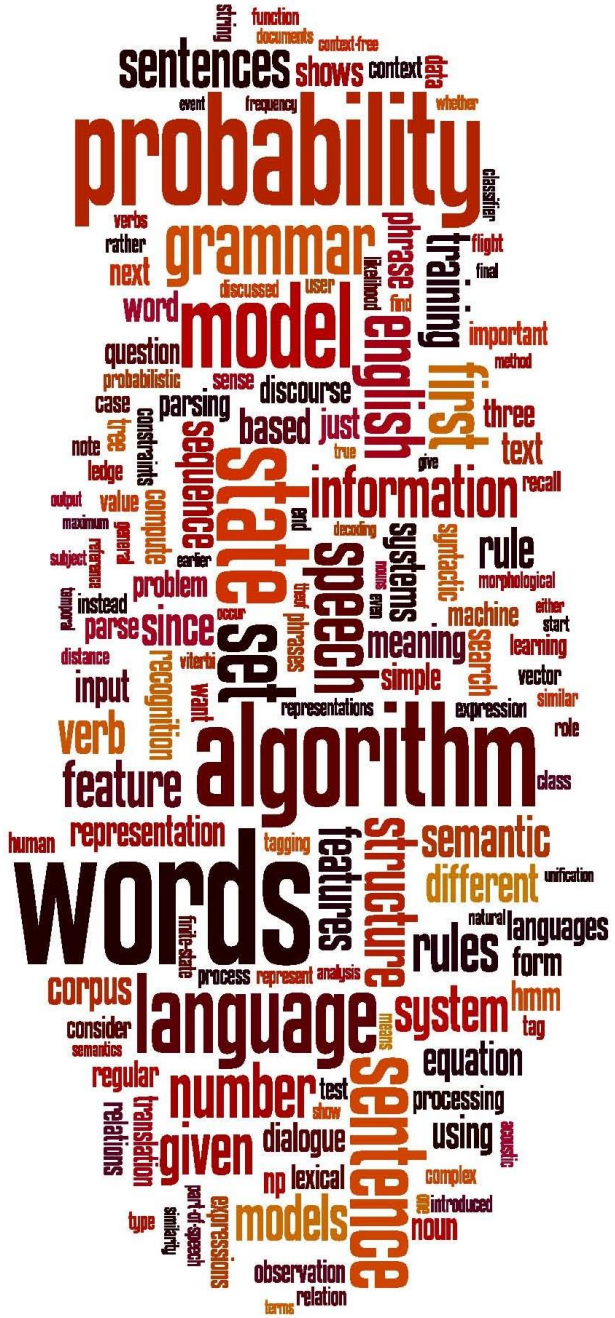
1. Does '*fish*' occur in window? - Yes. => bass1
2. Is the previous word '*striped* '? - Yes. => bass1
3. Does '*guitar*' occur in window? - Yes. => bass2
4. Is the following word '*player*'? - Yes. => bass2

Learning a decision list for a word with two senses:

- Define a **feature set**: what kind of questions do you want to ask?
- Enumerate all features (questions) the training data gives answers for
- Score each feature:

$$score(f_i) = \left|\log \frac{P(sense_1|f_i)}{P(sense_2|f_i)}\right|$$

-Rank all features by their score



Word Sense Disambiguation

- Evaluations and Baselines

WSD EVALUATIONS AND BASELINES

- Best evaluation: **extrinsic (‘end–to–end’, `task--based’) evaluation**
  - Embed WSD algorithm in a task and see if you can do the task better!
- What we often do for convenience: **intrinsic evaluation**
  - Exact match **sense accuracy**
    - % of words tagged identically with the human-manual sense tags
  - Usually evaluate using **held–out data** from same labeled corpus
- Baselines
  - Most frequent sense
  - The Lesk algorithm

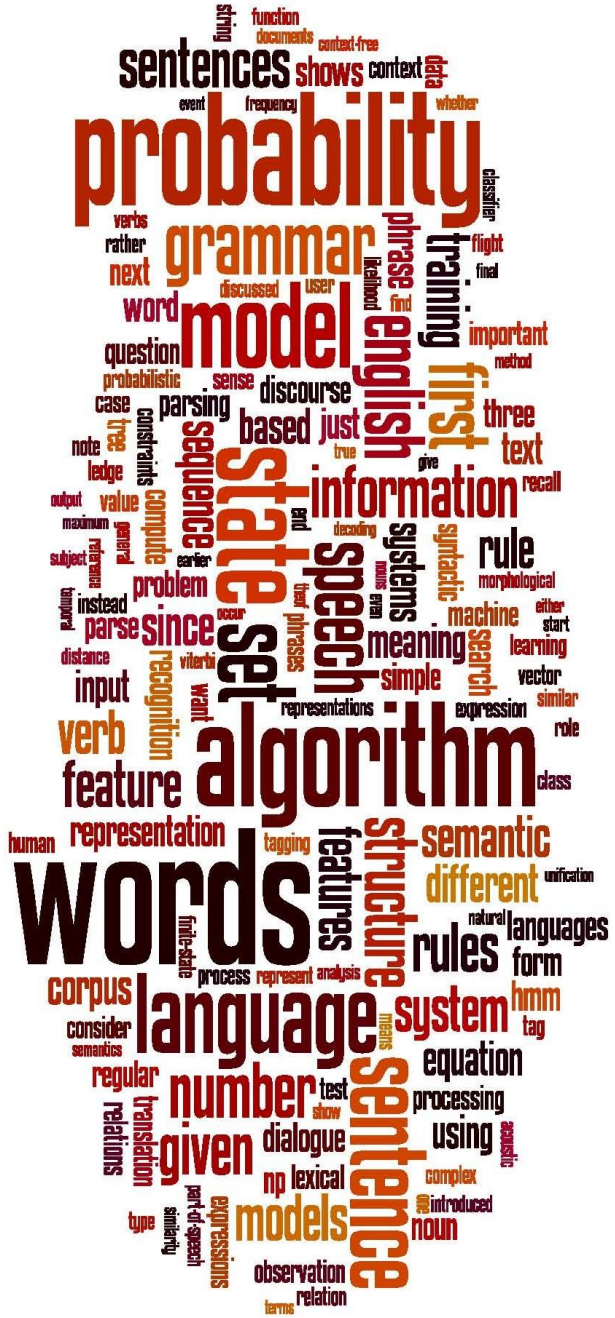
MOST FREQUENT SENSE

- WordNet senses are ordered in frequency order
- So "most frequent sense" in WordNet = "take the first sense"
- Sense frequencies come from the *SemCor* corpus

Freq	Synset	Gloss
338	plant[1], works, industrial plant	buildings for carrying on industrial labor
207	plant[2], flora, plant life	a living organism lacking the power of locomotion
2	plant[3]	something planted secretly for discovery by another
0	plant[4]	an actor situated in the audience whose acting is rehearsed but seems spontaneous to the audience

CEILING

- Human inter-annotator agreement
  - Compare annotations of two humans
  - On same data
  - Given same tagging guidelines
- Human agreements on all-words corpora with WordNet style senses
  - 75%–80%



Word Sense Disambiguation

- Dictionary and Thesaurus Methods

Dictionary-based approaches

- Rely on machine readable dictionaries
- Initial implementation of this kind of approach is due to Michael Lesk (1986)
- "Lesk algorithm"
  - Given a word *W* to be disambiguated in context C
    - Retrieve all of the sense definitions, *S*, for *W* from the MRD
    - Compare each *s* in *S* to the dictionary definitions D of all the remaining words c in the context C
    - Select the sense *s* with the most overlap with D (the definitions of the context words C)

Simplified Lesk Algorithm

```
function SIMPLIFIED LESK(word, sentence) returns best sense of word

  best-sense ← most frequent sense for word
  max-overlap ← 0
  context ← set of words in sentence
  for each sense in senses of word do
    signature ← set of words in the gloss and examples of sense
    overlap ← COMPUTEOVERLAP(signature, context)
    if overlap > max-overlap then
        max-overlap ← overlap
        best-sense ← sense
  end
  return(best-sense)
```

The Simplified Lesk algorithm

- Let's disambiguate "**bank**" in this sentence:

  The **bank** can guarantee deposits will eventually cover future tuition costs because it invests in adjustable-rate mortgage securities.

- given the following two WordNet senses:

bank[1]	Gloss:	a financial institution that accepts deposits and channels the money into lending activities
	Examples:	"he cashed a check at the bank", "that bank holds the mortgage on my home"
bank[2]	Gloss:	sloping land (especially the slope beside a body of water)
	Examples:	"they pulled the canoe up on the bank", "he sat on the bank of the river and watched the currents"

The Simplified Lesk algorithm

Choose sense with most word overlap between gloss and context
(not counting function words)

The **bank** can guarantee deposits will eventually cover future tuition costs because it invests in adjustable-rate mortgage securities.

bank[1]	Gloss:	a financial institution that accepts deposits and channels the money into lending activities
	Examples:	"he cashed a check at the bank", "that bank holds the mortgage on my home"
bank[2]	Gloss:	sloping land (especially the slope beside a body of water)
	Examples:	"they pulled the canoe up on the bank", "he sat on the bank of the river and watched the currents"

The Lesk algorithm

Basic idea: Compare the context with the dictionary definition of the sense.

Assign the dictionary sense whose gloss and examples are most similar to the context in which the word occurs.

Compare the **signature of a word** **in context**
with the **signatures of its senses** **in the dictionary**
Assign the sense that is **most similar** to the context

Signature = set of content words
(in examples/gloss or in context)
Similarity = size of intersection of context signature and sense signature

Sense signatures (dictionary)

***bank1*:**

Gloss: a financial institution that accepts deposits and channels the money into lending activities

Examples: *"he cashed the check at the bank", "that bank holds the mortgage on my home"*

***Signature(bank1)* = {*financial, institution, accept, deposit, channel, money, lend, activity, cash, check, hold, mortgage, home*}**

***bank2*:**

Gloss: sloping land (especially the slope beside a body of water)

Examples: *"they pulled the canoe up on the bank", "he sat on the bank of the river and watched the current"*

***Signature(bank2)* = {*slope, land, body, water, pull, canoe, sit, river, watch, current*}**

Signature of target word

Test sentence:
*"The **bank** refused to give me a loan."*

Simplified Lesk: Overlap between sense signature and (simple) signature of the target word:

Target signature = words in context: **{refuse, give, loan}**

Original Lesk: Overlap between sense signature and augmented signature of the target word

Augmented target signature with signatures of words in context **{refuse**, reject, request,... , **give**, gift, donate,... **loan,** money, borrow,...}

PROS AND CONS

- Pros
  - Simple
  - Does not require (human-annotated) training data

- Cons
  - Very sensitive to the definition of words
  - Words used in definition might not overlap with the context.
  - Even if there is a human annotated training data, it does not learn from the data.

THE CORPUS LESK ALGORITHM

- Assumes we have some sense--labeled data (like SemCor)
- Take all the sentences with the relevant word sense:

  These short, "streamlined" meetings usually are sponsored by local ***banks[1]****, Chambers of Commerce, trade associations, or other civic organizations.*

- Now add these to the gloss + examples for each sense, call it the "signature" of a sense.
- Choose sense with most word overlap between context and signature.

CORPUS LESK: IDF WEIGHTING

- Instead of just removing function words
  - Weigh each word by its `promiscuity' across documents
  - Down--weights words that occur in every `document' (gloss, example, etc)
  - These are generally function words, but is a more fine--grained measure
- Weigh each overlapping word by **inverse document frequency**

CORPUS LESK: IDF WEIGHTING

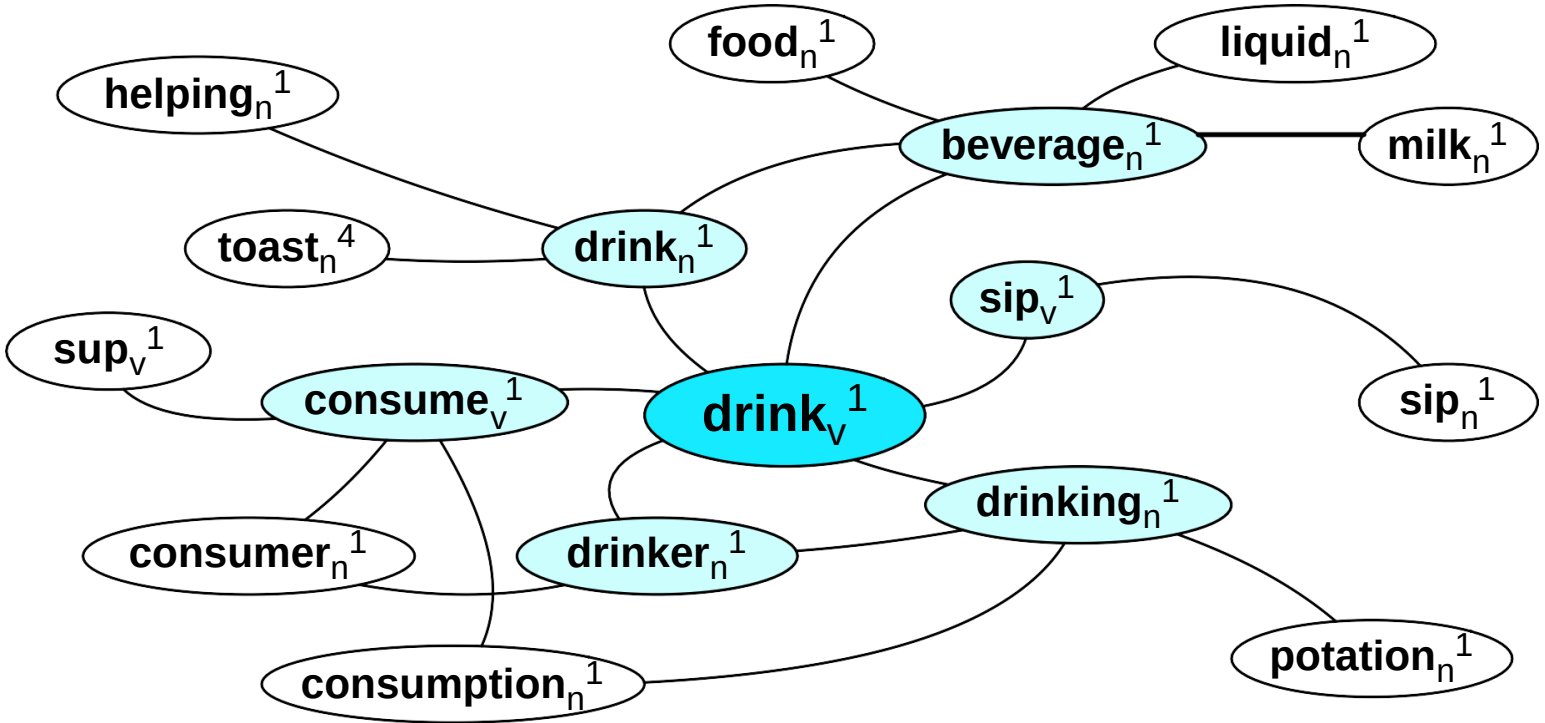
- Weigh each overlapping word by **inverse document frequency**
  - N is the total number of documents
  - df_i = "document frequency of word *i*"
  - = # of documents with word *I*

$$\text{idf}_i = \log\left(\frac{N}{df_i}\right)$$

$$score(sense_i, context_j) = \sum_{w \in overlap(signature_i, context_j)} \text{idf}_w$$

GRAPH--BASED METHODS

- First, WordNet can be viewed as a graph
  - senses are nodes
  - relations (hypernymy, meronymy) are edges
  - Also add edge between word and unambiguous gloss words



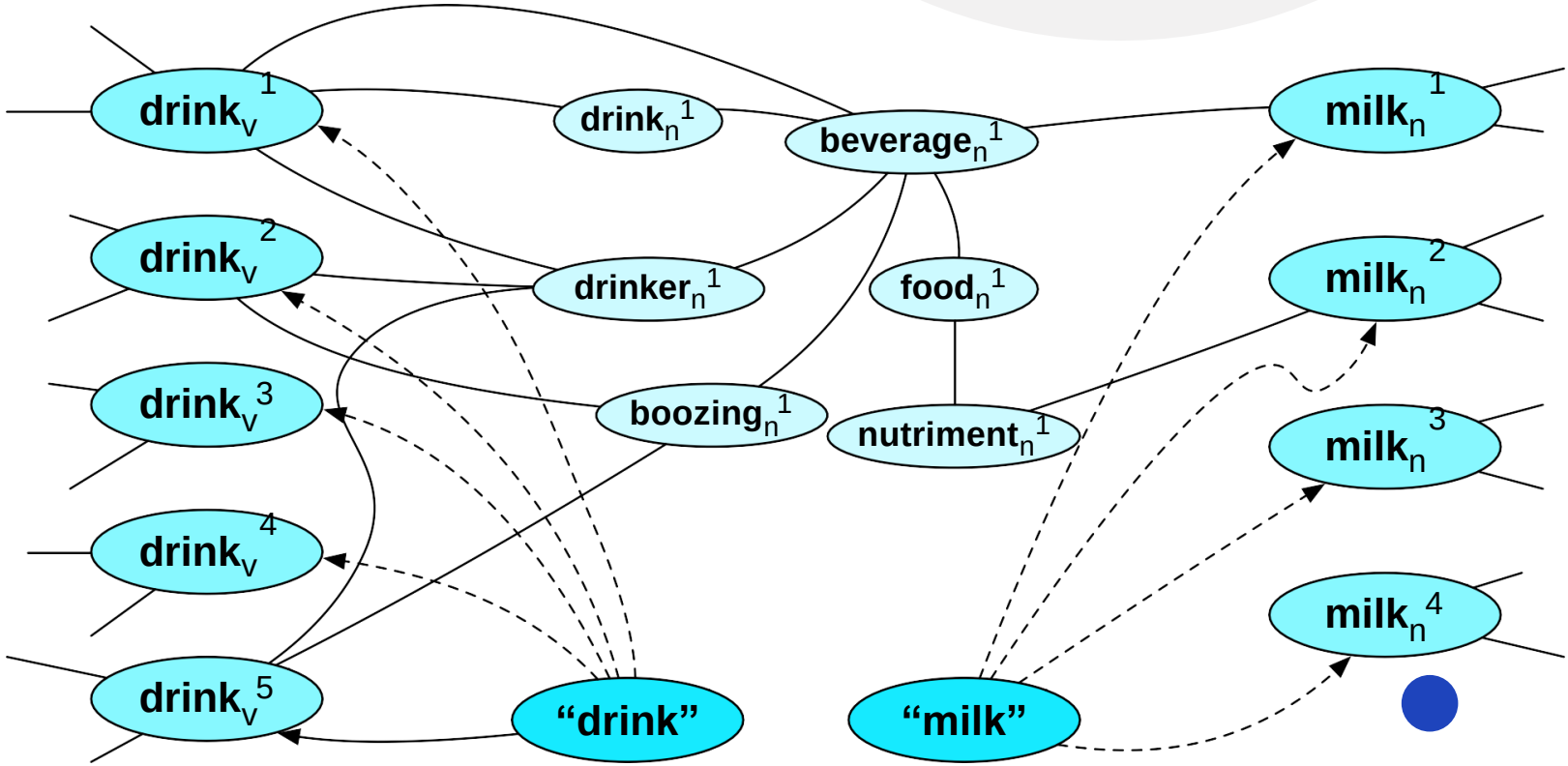
HOW TO USE THE GRAPH FOR WSD

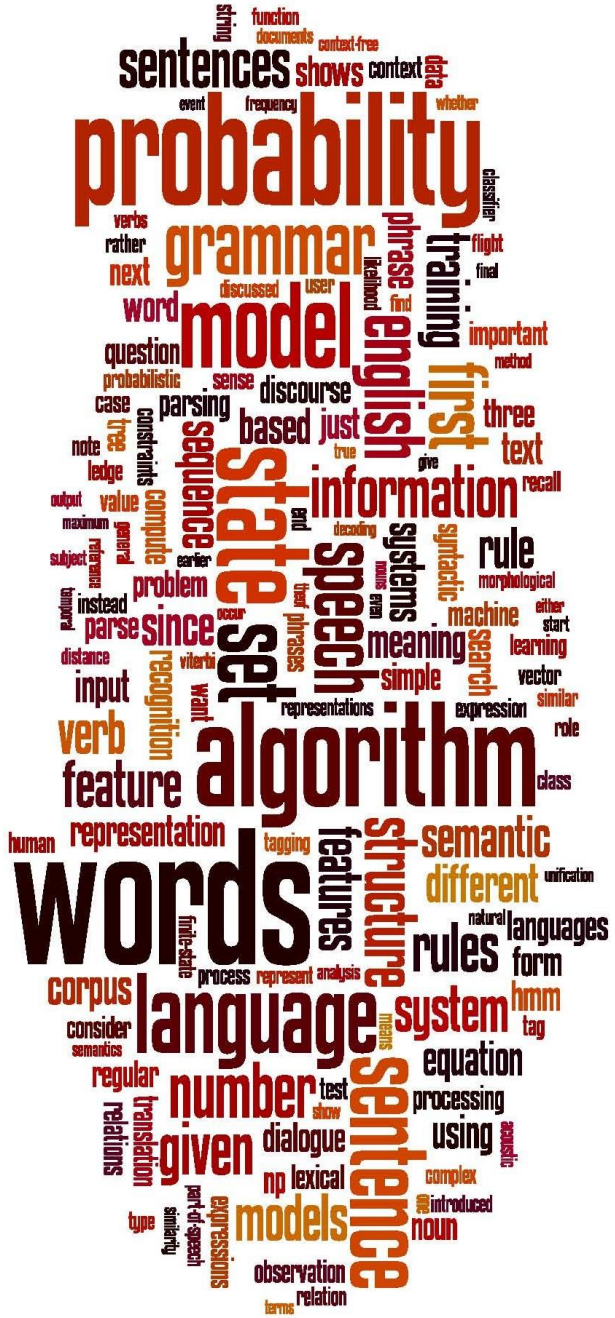
- Insert target word and words in its sentential context into the graph, with directed edges to their senses

"She drank some milk"

- Now choose the *most central* sense

Add some probability to "drink" and "milk" and compute node with highest "pagerank"





Word Sense Disambiguation

Semi-Supervised Learning

SEMI--SUPERVISED LEARNING

PROBLEM: Supervised and dictionary--based approaches require large hand-built resources

What if you don't have so much training data?

Solution: Bootstrapping

Generalize from a very small hand--labeled seed--set.

BOOTSTRAPPING

- For `bass`
  - Rely on “One sense per collocation” rule
    - A word reoccurring in collocation with the same word will almost surely have the same sense.

  - the word `play` occurs with the music sense of bass
  - the word `fish` occurs with the fish sense of bass

SENTENCES EXTRACTING USING “FISH” AND “PLAY”

We need more good teachers – right now, there are only a half a dozen who can **play** the free **bass** with ease. An electric guitar and **bass play**er stand off to one side, not really part of the scene, just as a sort of nod to gringo expectations perhaps.
The researchers said the worms spend part of their life cycle in such **fish** as Pacific salmon and striped **bass** and Pacific rockfish or snapper. And it all started when **fish**ermen decided the striped **bass** in Lake Mead were too skinny.

SEMI-SUPERVISED: YAROWSKY ALGORITHM

The task:

Learn a **decision list classifier** for each ambiguous word (e.g. "*plant": living/factory?*) from lots of **unlabeled sentences**.

Features used by the classifier:

- **Collocations:** *"plant life", "manufacturing plant"*
- **Nearby (± 2-10) words:** *"animal ", "automate"*

Assumption 1: **One-sense-per-collocation**

"plant" in *"plant life"* always refers to *living* plants

Assumption 2: One-sense-per-discourse

A text talks either about living plants or about factories.

YAROWSKY'S TRAINING REGIME

1. **Initialization:**
   - Label a few seed examples.
   - Train an initial classifier on these seed examples
2. **Relabel:**
   - Label all examples with current classifier.
   - Put all examples that are labeled with high confidence into a new labeled data set.
   - Optional: apply one-sense-per-discourse to correct mistakes and get additional labels
3. **Retrain:**
   - Train a new classifier on the new labeled data set.
4. Repeat 2. and 3. until convergence.

INITIAL STATE: FEW LABELS

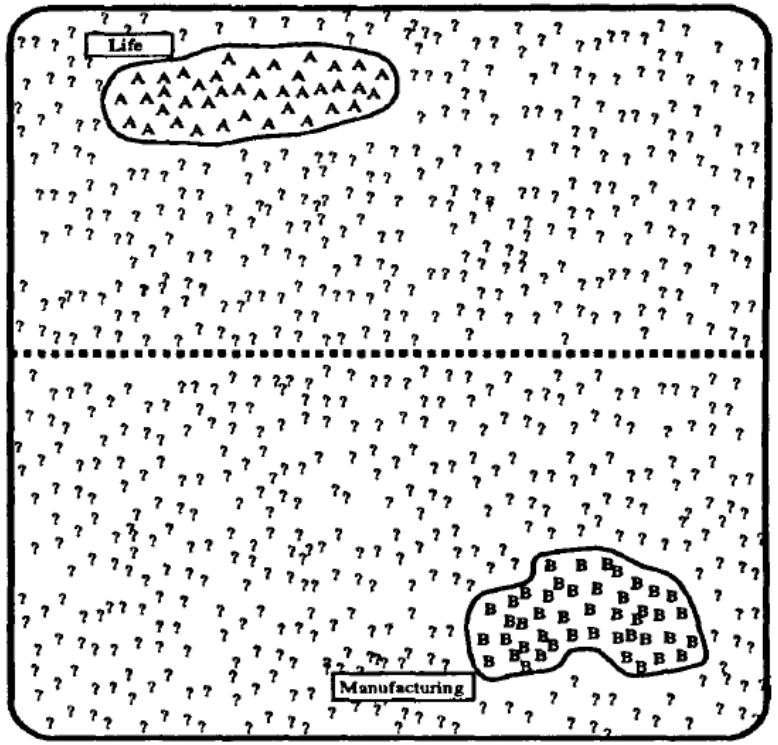


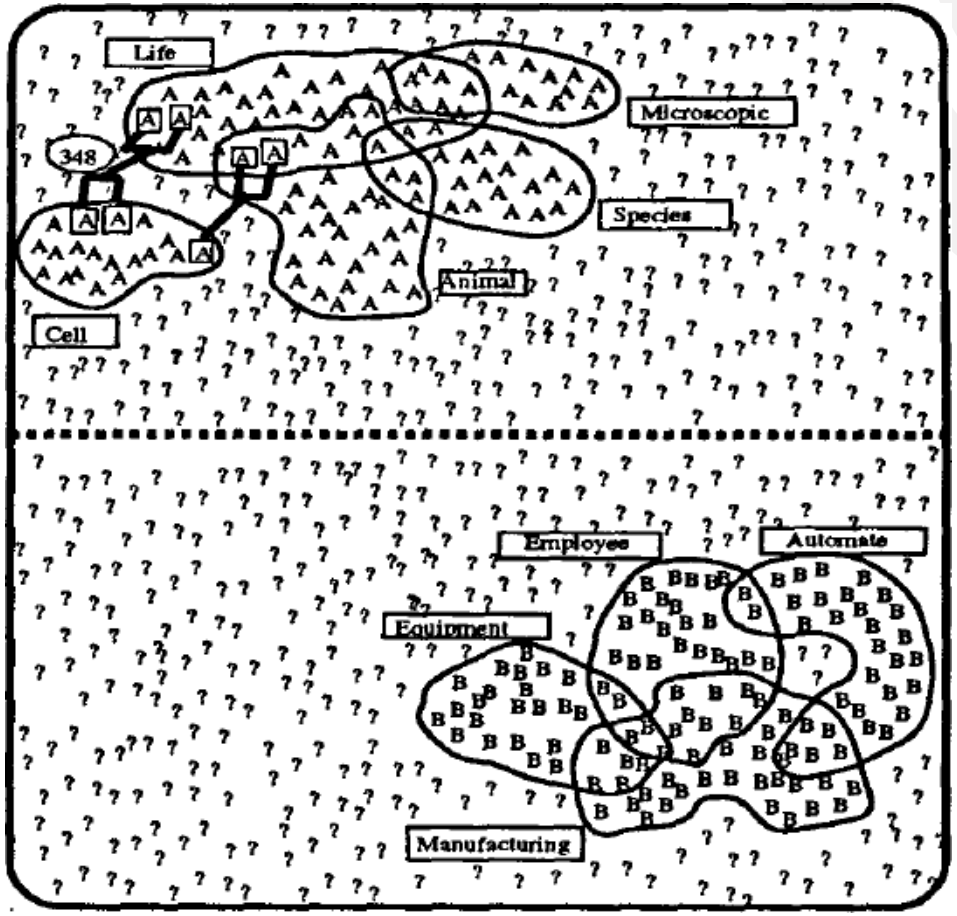
Figure 1: Sample Initial State

A = SENSE-A training example
B = SENSE-B training example
? = currently unclassified training example
Life = Set of training examples containing the collocation "life".

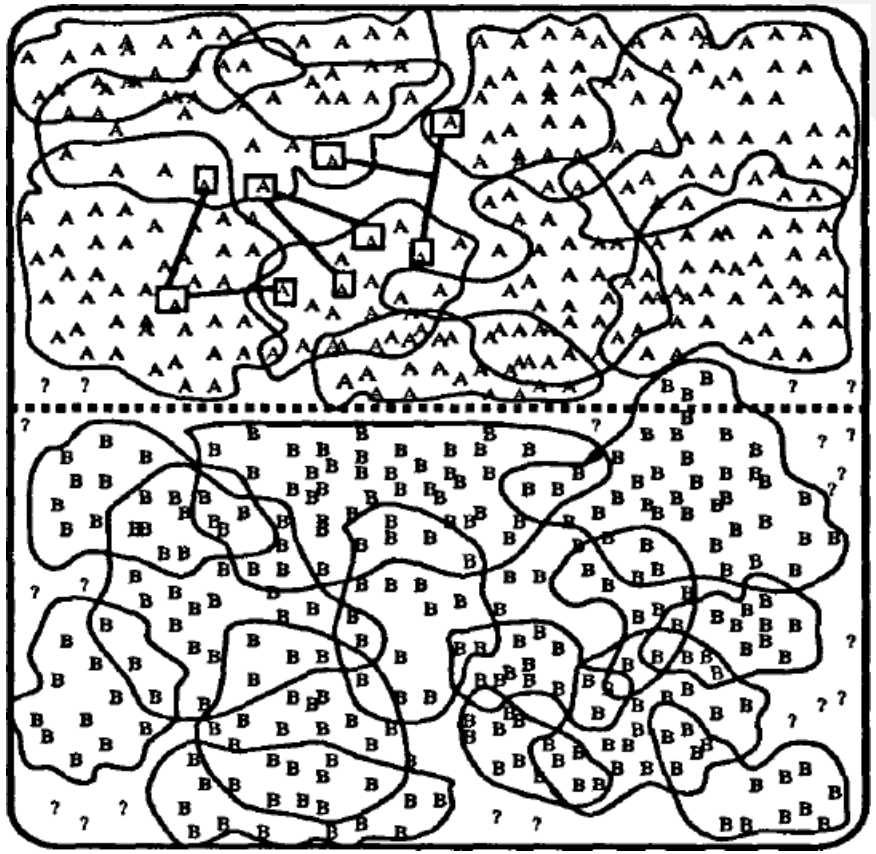
THE INITIAL DECISION LIST

Initial decision list for *plant* (abbreviated)		
LogL	Collocation	Sense
8.10	***plant* life**	⇒ A
7.58	**manufacturing *plant***	⇒ B
7.39	**life** (within ±2-10 words)	⇒ A
7.20	**manufacturing** (in ±2-10 words)	⇒ B
6.27	animal (within ±2-10 words)	⇒ A
4.70	equipment (within ±2-10 words)	⇒ B
4.39	employee (within ±2-10 words)	⇒ B
4.30	assembly *plant*	⇒ B
4.10	*plant* closure	⇒ B
3.52	*plant* species	⇒ A
3.48	automate (within ±2-10 words)	⇒ B
3.45	microscopic *plant*	⇒ A
	...	

INTERMEDIATE STATE: MORE LABELS



FINAL STATE: ALMOST EVERYTHING LABELED

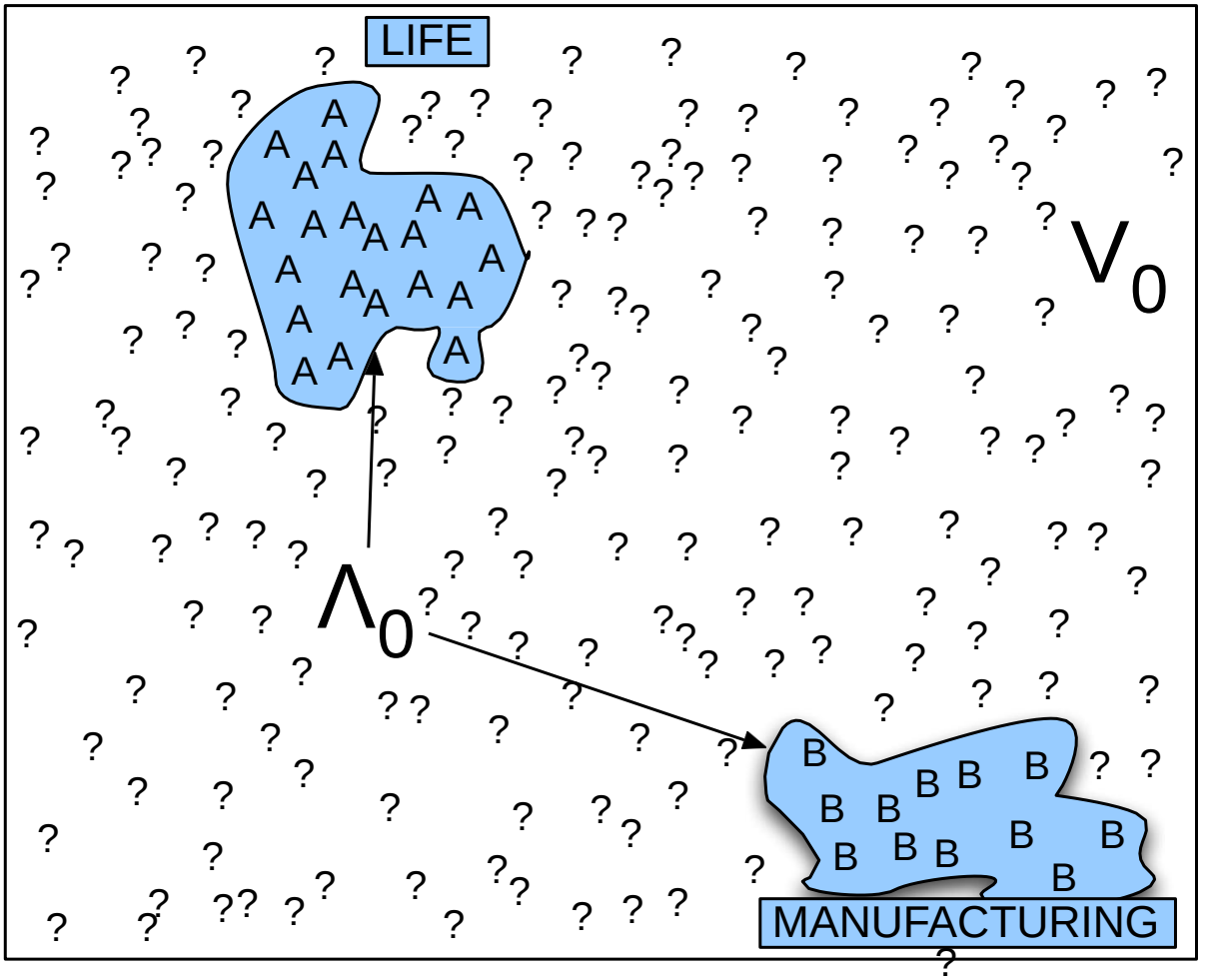


INITIAL VS. FINAL DECISION LISTS

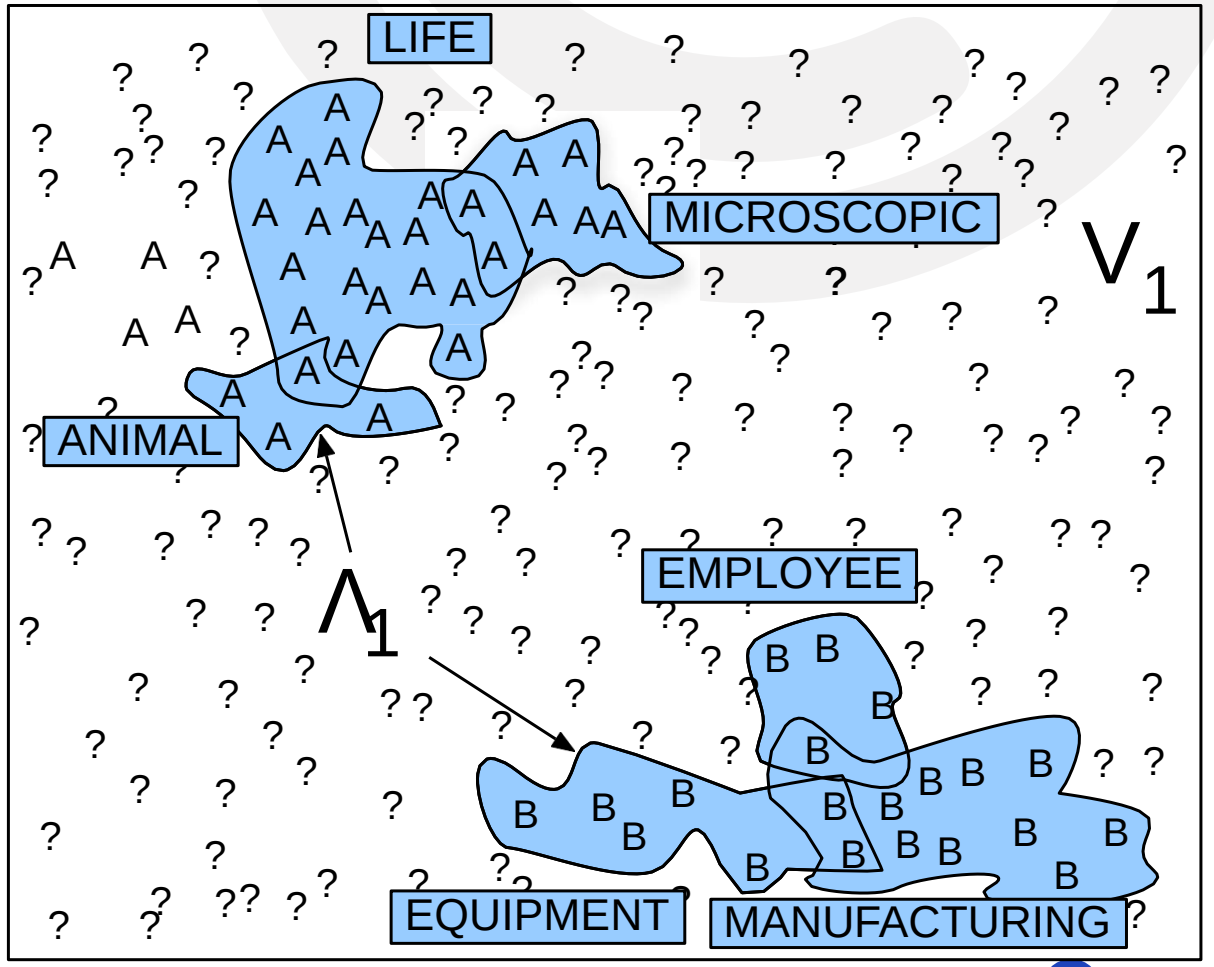
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3.52	*plant* species	⇒ A
3.48	automate (within ±2-10 words)	⇒ B
3.45	microscopic *plant*	⇒ A
	...	

Final decision list for *plant* (abbreviated)		
LogL	Collocation	Sense
10.12	*plant* growth	⇒ A
9.68	car (within ±*k* words)	⇒ B
9.64	*plant* height	⇒ A
9.61	union (within ±*k* words)	⇒ B
9.54	equipment (within ±*k* words)	⇒ B
9.51	assembly *plant*	⇒ B
9.50	nuclear *plant*	⇒ B
9.31	flower (within ±*k* words)	⇒ A
9.24	job (within ±*k* words)	⇒ B
9.03	fruit (within ±*k* words)	⇒ A
9.02	*plant* species	⇒ A
...	...	

STAGES IN THE YAROWSKY BOOTSTRAPPING ALGORITHM FOR THE WORD "PLANT"



(a)



(b)

SUMMARY: GENERATING SEEDS

1) Hand labeling
2) “One sense per collocation”:
   - A word reoccurring in collocation with the same word will almost surely have the same sense.
3) “One sense per discourse”:
   - The sense of a word is highly consistent within a document -Yarowsky (1995)
   - (At least for non-function words, and especially topic-specific words)

HYPERLEX

- **KEY IDEA**
  - Instead of using *"dictionary defined senses"* extract the "*senses from the corpus*" itself
  - These "*corpus senses*" or "*uses*" correspond to clusters of similar contexts for a word.

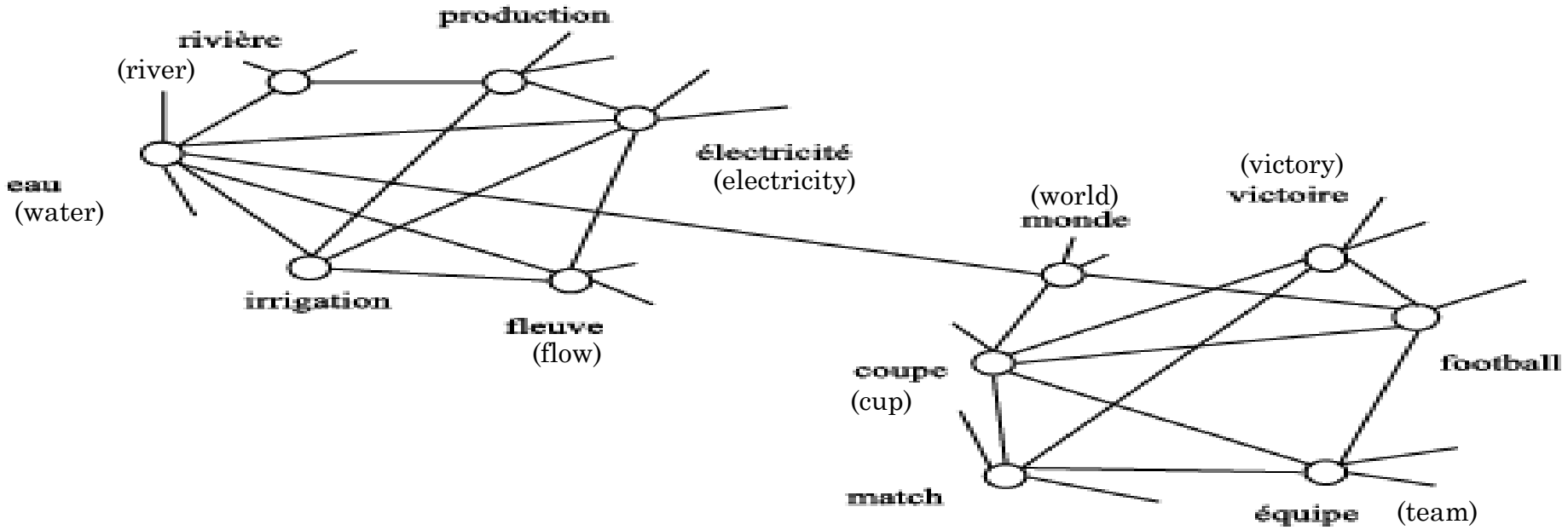
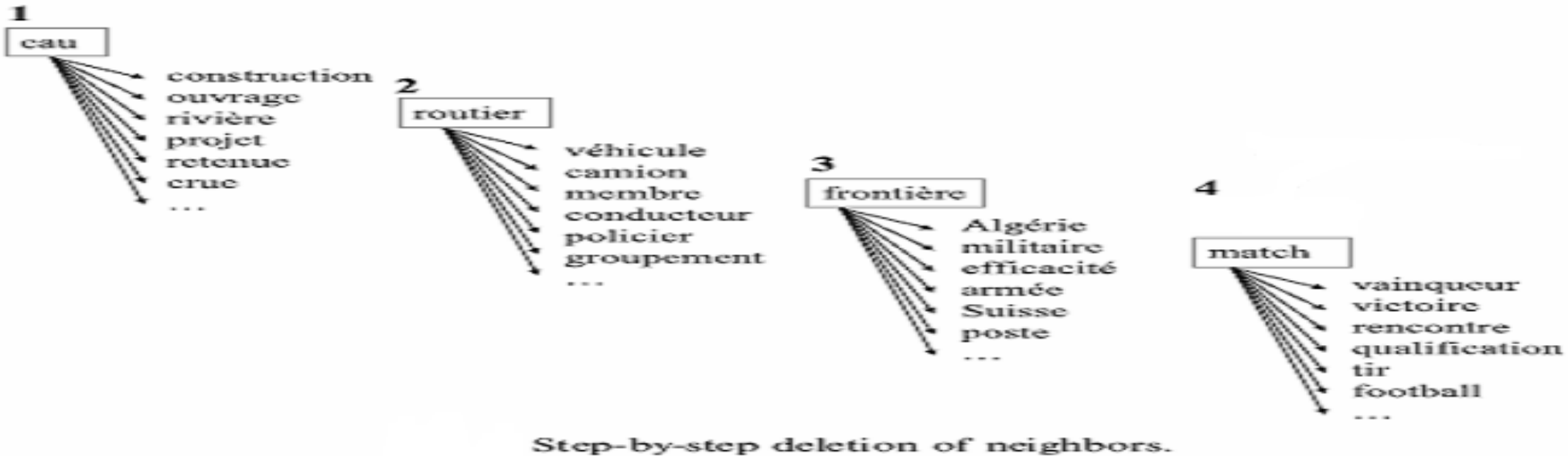


Fig. 4. Graph of the cooccurrents of the French word *barrage*.

Detecting Root Hubs

- Different uses of a target word form highly interconnected bundles (or high density components)
- In each high density component one of the nodes *(hub)* has a higher degree than the others.
- **Step 1:** Construct co-occurrence graph, G.
- **Step 2:** Arrange nodes in G in decreasing order of in-degree.
- **Step 3:** Select the node from G which has the highest frequency. This node will be the hub of the first high density component**.**
- **Step 4:**Delete this hub and all its neighbors from G.
- **Step 5:**Repeat Step 3 and 4 to detect the hubs of other high density components

Detecting Root Hubs (Contd.)



Step-by-step deletion of neighbors.

The four components for "barrage" can be characterized as:

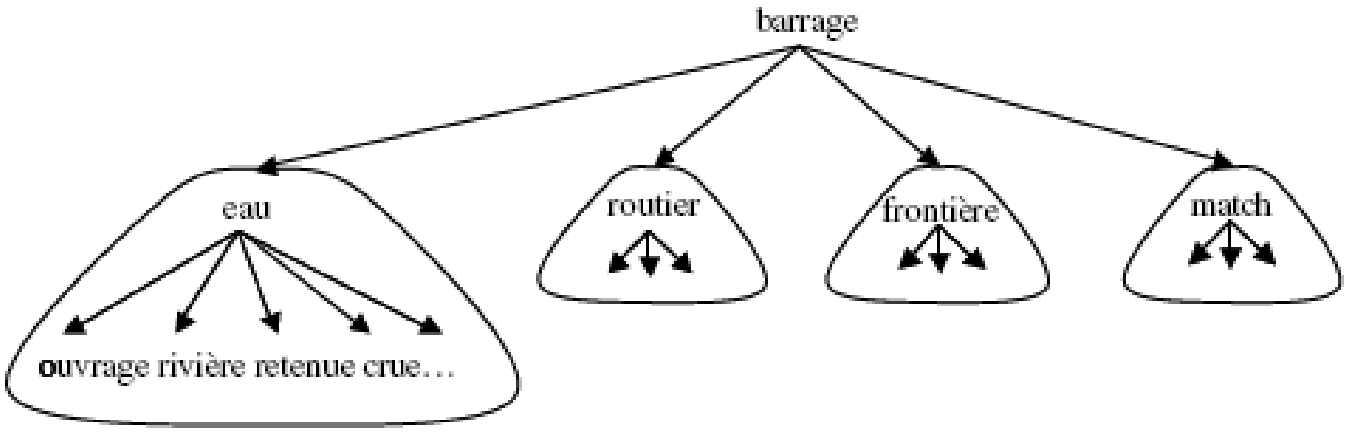
EAU	*construction ouvrage rivière projet retenue crue* (construction engineering-work river project reservoir flood)
ROUTIER	*véhicule camion membre conducteur policier groupement* (vehicle truck member driver policeman group)
FRONTIERE	*Algérie militaire efficacité armée Suisse poste* (Algeria military efficiency army Switzerland post)
MATCH	*vainqueur victoire rencontre qualification tir football* (winner victory encounter qualification shot soccer)

Delineating Components

- Attach each node to the root hub closest to it.
- The distance between two nodes is measured as the smallest sum of the weights of the edges on the paths linking them.
- **Step 1**:

  Add the target word to the graph G.

- **Step 2**:
  - Compute a *Minimum Spanning Tree (MST)* over G taking the target word as the root.



Minimum spanning tree and high-density components.

Disambiguation

- Each node in the MST is assigned a score vector with as many dimensions as there are components.

$$\mathbf{s}_i = \frac{1}{1 + d(h_i, v)} \quad \text{if } v \text{ belongs to component } i,$$

$$\mathbf{s}_i = 0 \quad \text{otherwise.}$$

$d(h_i, v)$ is the distance between root hub h_i and node v in the tree.

E.g. pluei(rain) belongs to the component EAU(water) and d(eau, pluie) = 0.82, s_{pluei} = (0.55, 0, 0, 0)

- **Step 1:** For a given context, add the score vectors of all words in that context.
- **Step 2:** Select the component that receives the highest weight.

Disambiguation (Example)

Le **barrage** recueille l'eau a la saison des plueis.

The **dam** collects water during the rainy season.

Scores for the context "Le barrage recueille l'eau à la saison des pluies"

	EAU	ROUTIER	FRONTIERE	MATCH
S_{eau}	1.00	0.00	0.00	0.00
S_{saison}	0.00	0.00	0.00	0.39
S_{pluie}	0.55	0.00	0.00	0.00
Total	1.55	0.00	0.00	0.39

A reliability coefficient (ρ) can be calculated as the difference (δ) between the best score and the second best score.

$$\rho = 1 - (1/(1+ \delta))$$

SUMMARY

- Word Sense Disambiguation: choosing correct sense in context
- Applications: MT, QA, etc.
- Three classes of Methods
  - Supervised Machine Learning: Naive Bayes classifier
  - Thesaurus/Dictionary Methods
  - Semi-Supervised Learning
- Main intuition
  - There is lots of information in a word's context
  - Simple algorithms based just on word counts can be surprisingly good